

Student Utility Program — Java Lab Record

1. Aim

To develop a menu-driven Java Student Utility Program that accepts student details and marks, calculates the percentage and grade, and performs various operations using conditional statements, switch-case, loops, methods, and user input.

2. Step-by-Step Procedure

1. Create a Java class named `StudentUtility`.
2. Import the `Scanner` class to read input from the user.
3. Define methods for factorial, prime checking, maximum of two numbers, and area of a circle.
4. Display the student information heading and accept the student's name and roll number.
5. Read marks of five subjects and calculate the total marks.
6. Calculate the percentage and display the appropriate grade.
7. Display the menu containing Even/Odd, Largest of Three, Day of Week, Multiplication Table, Numbers from 1 to N, Sum of Natural Numbers, Fibonacci Series, Methods, and Exit.
8. Read the user's menu choice using `Scanner`.
9. Use switch-case to execute the selected operation.
10. Use a do-while loop to repeatedly display the menu until the user selects Exit.
11. Close the `Scanner` object after termination.

3. Algorithm

1. Start.
2. Create a `Scanner` object.
3. Display STUDENT INFORMATION.
4. Read student name and roll number.
5. Read marks of five subjects and calculate total marks.
6. Calculate $\text{percentage} = \text{total marks} / 5.0$.
7. Display student details, percentage, and grade.
8. Display the utility menu.
9. Read the user's choice.
10. If choice is 1, check whether the number is even or odd.
11. If choice is 2, find the largest of three numbers.
12. If choice is 3, display the day corresponding to the day number.
13. If choice is 4, display the multiplication table.
14. If choice is 5, display numbers from 1 to N.
15. If choice is 6, calculate the sum of the first N natural numbers.

16. If choice is 7, display the Fibonacci series.
17. If choice is 8, execute the methods for factorial, prime checking, maximum, and circle area.
18. If choice is 9, display Exit.
19. Otherwise, display Invalid Choice.
20. Repeat the menu until the choice is 9.
21. Close Scanner.
22. Stop.

4. UML Class Diagram

```
+-----+
| StudentUtility                                |
+-----+
| + factorial(n : int) : int                    |
| + isPrime(n : int) : boolean                 |
| + maximum(a : int, b : int) : int             |
| + areaCircle(r : double) : double            |
| + main(args : String[]) : void               |
+-----+
| Static utility methods and main method      |
+-----+

<>
|
v
+-----+
| Scanner                                      |
+-----+
| next()                                       |
| nextInt()                                   |
| nextDouble()                               |
+-----+

Main operations:
Student Details → Grade → Menu → Selected Operation
```

5. Code

```
package core_java;
import java.util.Scanner;

public class StudentUtility {
    // Factorial Method
    static int factorial(int n) {
        int fact = 1;
        for (int i = 1; i < n; i++) {
            fact = fact * i;
        }
    }
}
```

```

        return fact;
    }

    // Prime Method
    static boolean isPrime(int n) {
        if (n <= 1) return false;
        for (int i = 2; i <= n/2; i++) {
            if (n % i == 0) return false;
        }
        return true;
    }

    // Maximum Method
    static int maximum(int a, int b) {
        if (a > b) return a;
        else return b;
    }

    // Area of Circle Method
    static double areaCircle(double r) {
        double A = 3.14 * r * r;
        return A;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Part A: Student Details
        System.out.println(" STUDENT INFORMATION \n");
        System.out.print("Enter Student Name: ");
        String name = sc.next();
        System.out.print("Enter Roll Number: ");
        int roll = sc.nextInt();
        int total = 0;
        System.out.println("Enter Marks of 5 Subjects:");
        for (int i = 1; i <= 5; i++) {
            System.out.print("Subject " + i + ": ");
            total = total + sc.nextInt();
        }
        double percentage = total / 5.0;
        System.out.println("\nStudent Name: " + name);
        System.out.println("Roll Number : " + roll);
        System.out.println("Total Marks : " + total);
        System.out.println("Percentage : " + percentage);

        // Grade
        if (percentage >= 90) System.out.println("Grade: A+");
        else if (percentage >= 80) System.out.println("Grade: A");
        else if (percentage >= 70) System.out.println("Grade: B");
        else if (percentage >= 60) System.out.println("Grade: C");
    }
}

```

```

else System.out.println("Grade: Fail");

int choice;
do {
    System.out.println("1. Even or Odd");
    System.out.println("2. Largest of Three Numbers");
    System.out.println("3. Day of Week");
    System.out.println("4. Multiplication Table");
    System.out.println("5. Numbers from 1 to N");
    System.out.println("6. Sum of First N Natural Numbers");
    System.out.println("7. Fibonacci Series");
    System.out.println("8. Methods");
    System.out.println("9. Exit");
    System.out.print("Enter your choice: ");
    choice = sc.nextInt();

    switch (choice) {
        case 1:
            System.out.print("Enter a number: ");
            int num = sc.nextInt();
            if (num % 2 == 0) System.out.println("Even Number");
            else System.out.println("Odd Number");
            break;
        case 2:
            System.out.print("Enter Three Numbers: ");
            int a = sc.nextInt();
            int b = sc.nextInt();
            int c = sc.nextInt();
            if (a > b && a > c) System.out.println("Largest Number = " +
a);

            else if (b > a && b > c) System.out.println("Largest Number =
" + b);

            else System.out.println("Largest Number = " + c);
            break;
        case 3:
            System.out.print("Enter Day Number (1-7): ");
            int day = sc.nextInt();
            switch (day) {
                case 1: System.out.println("Monday"); break;
                case 2: System.out.println("Tuesday"); break;
                case 3: System.out.println("Wednesday"); break;
                case 4: System.out.println("Thursday"); break;
                case 5: System.out.println("Friday"); break;
                case 6: System.out.println("Saturday"); break;
                case 7: System.out.println("Sunday"); break;
                default: System.out.println("Invalid Day");
            }
            break;
        case 4:
            System.out.print("Enter Number: ");

```

```

        int table = sc.nextInt();
        for (int i = 1; i <= 10; i++) {
            System.out.println(table + " x " + i + " = " + (table *
i));
        }
        break;
    case 5:
        System.out.print("Enter N: ");
        int n = sc.nextInt();
        for (int i = 1; i <= n; i++) {
            System.out.print(i + " ");
        }
        System.out.println();
        break;
    case 6:
        System.out.print("Enter N: ");
        int num1 = sc.nextInt();
        int sum = 0;
        for (int i = 1; i <= num1; i++) {
            sum = sum + i;
        }
        System.out.println("Sum = " + sum);
        break;
    case 7:
        System.out.print("Enter Number of Terms: ");
        int terms = sc.nextInt();
        int x = 0, y = 1;
        System.out.print("Fibonacci Series: ");
        for (int i = 1; i <= terms; i++) {
            System.out.print(x + " ");
            int next = x + y;
            x = y;
            y = next;
        }
        System.out.println();
        break;
    case 8:
        System.out.print("Enter Number for Factorial: ");
        int f = sc.nextInt();
        System.out.println("Factorial = " + factorial(f));
        System.out.print("Enter Number to Check Prime: ");
        int p = sc.nextInt();
        if (isPrime(p)) System.out.println("Prime Number");
        else System.out.println("Not Prime");
        System.out.print("Enter Two Numbers: ");
        int m1 = sc.nextInt();
        int m2 = sc.nextInt();
        System.out.println("Maximum = " + maximum(m1, m2));
        System.out.print("Enter Radius: ");
        double r = sc.nextDouble();

```

```

        System.out.println("Area = " + areaCircle(r));
        break;
    case 9:
        System.out.println("Exit");
        break;
    default:
        System.out.println("Invalid Choice");
    }
} while (choice != 9);
sc.close();
}
}

```

6. Sample Output

Below are rendered PowerShell terminal execution screenshots for the Student Utility Program:

```

Windows PowerShell - Output Screenshot 1

PS C:\Users\VYSHNAVI\OneDrive\Desktop\megh> javac StudentUtility.java
PS C:\Users\VYSHNAVI\OneDrive\Desktop\megh> java StudentUtility
STUDENT INFORMATION

Enter Student Name: vyshu
Enter Roll Number: 166
Enter Marks of 5 Subjects:
Subject 1: 98
Subject 2: 97
Subject 3: 100
Subject 4: 94
Subject 5: 96

Student Name : vyshu
Roll Number  : 166
Total Marks  : 485
Percentage   : 97.0
Grade : A+

1. Even or Odd
2. Largest of Three Numbers
3. Day of Week
4. Multiplication Table
5. Numbers from 1 to N
6. Sum of First N Natural Numbers
7. Fibonacci Series
8. Methods
9. Exit
Enter your choice: 1
Enter a number: 3
Odd Number

```

1. Even or Odd
2. Largest of Three Numbers
3. Day of Week
4. Multiplication Table
5. Numbers from 1 to N
6. Sum of First N Natural Numbers
7. Fibonacci Series
8. Methods
9. Exit

Enter your choice: 2

Enter Three Numbers: 98

97

34

Largest Number = 98

1. Even or Odd
2. Largest of Three Numbers
3. Day of Week
4. Multiplication Table
5. Numbers from 1 to N
6. Sum of First N Natural Numbers
7. Fibonacci Series
8. Methods
9. Exit

Enter your choice: 3

Enter Day Number (1-7): 5

Friday